SpaceX Falcon 9 landing analysis

Data Science Capstone Project Report

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https://github.com/Zeppelin567/ibm-spacex-capstone

IBM Skills Network course snapshots, accessed 21 Oct 2026

Executive summary

We analyzed API and Wikipedia snapshots with pandas and SQL, explored launch outcomes through charts, Folium and Dash, and compared four classifiers.

The API sample contains 90 launches and 60 successful first-stage outcomes (66.7%). Landing rates improved over the period.

A decision tree led training cross-validation at 90.4%. Its held-out test accuracy was 83.3% on 18 rows. These results need validation on recent launches.

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Introduction

Reusable first stages can influence launch economics. This project examines which observed launch characteristics relate to successful Falcon 9 first-stage landing.

The task is to collect launch records, describe patterns and predict the landing class.

Landing success differs from mission success. The historical sample cannot directly establish launch cost or operational safety.

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Data sources and populations

API analysis: 90 Falcon 9 records through 13 Nov 2020, after removing multi-core and multi-payload records.

Web collection: 121 numeric flight rows extracted from the lab’s specified Wikipedia revision.

SQL analysis: the separate official 101-row SpaceX CSV.

Dashboard: the official 56-row historical dashboard CSV.

Each result identifies its population. We do not combine these denominators.

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Data collection with the SpaceX API

1. Request the launches/past endpoint and normalize the JSON. The first static-fire date is in 2006.

2. Keep one-core, one-payload launches through 13 Nov 2020.

3. Enrich rocket, launchpad, payload and core IDs with metadata.

4. Remove Falcon 1, leaving 90 Falcon 9 records. LandingPad has 26 missing values.

The report reproduces filtering from the frozen API response and uses the official enriched CSV.

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

Data collection with web scraping

Request the exact Wikipedia revision specified in the lab: oldid=1027686922.

BeautifulSoup selects wikitable launch tables, then numeric flight rows with at least nine data cells.

Extract date/time, booster, site, payload, orbit, customer and both outcomes. Strip citation markers and normalize Unicode whitespace.

The saved extraction contains 121 unique flight rows. Raw HTML and extracted CSV support reproduction.

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

Data wrangling methodology

Convert dates, inspect nulls and retain explicit missing pad IDs. The course enriched snapshot already replaces missing payload mass with its mean.

Assign class 1 to True ASDS, True RTLS and True Ocean. All other first-stage outcomes receive class 0.

Encode Orbit, LaunchSite, LandingPad and Serial with get_dummies. The course feature matrix has 80 columns.

For modeling, fit preprocessing inside training folds. The upstream payload imputation remains a limitation.

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

EDA with data visualization

Scatter plots compare flight sequence and payload mass across launch sites and orbits, with points colored by landing class.

Bar charts compare observed landing rates. A yearly line chart shows historical change.

The 90-row API sample contains 55 CCAFS SLC 40, 22 KSC LC 39A and 13 VAFB SLC 4E launches.

Small groups and changes in hardware over time limit site and orbit comparisons.

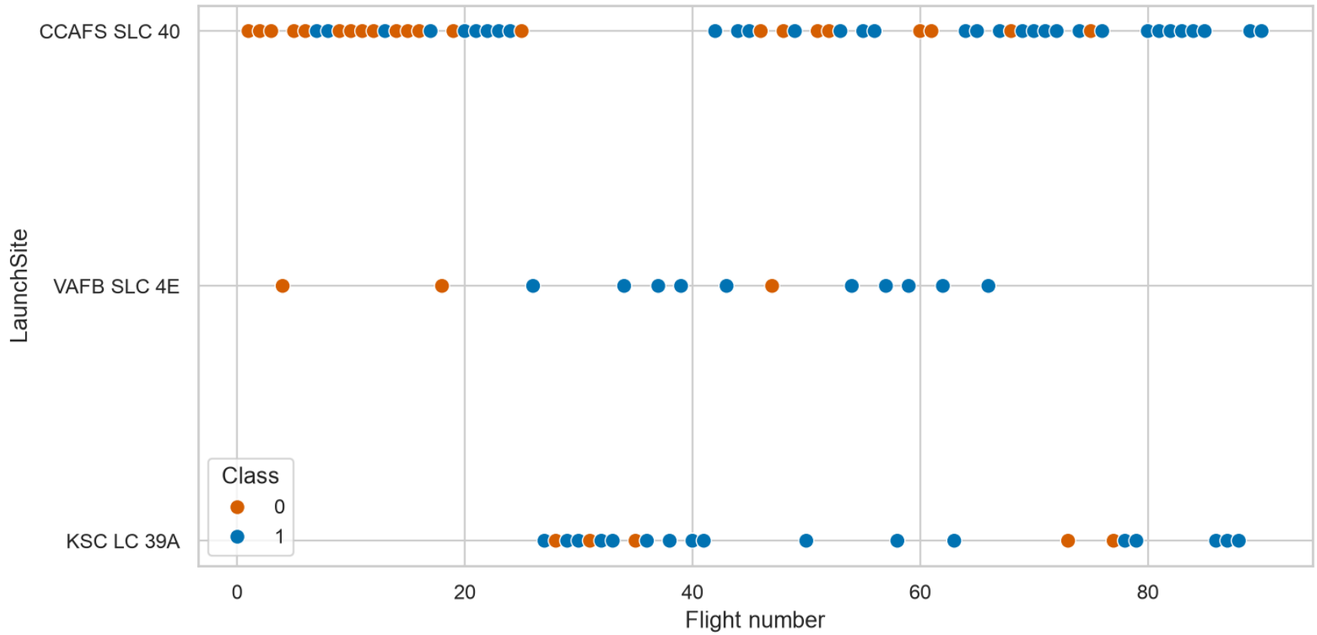
https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

Flight sequence and launch site

Successful landing outcomes become more common later in the sequence. This pattern can reflect improvements in hardware and operations.

Site codes: 1 = CCAFS SLC 40, 2 = KSC LC 39A, 3 = VAFB SLC 4E.

Blue = class 1. Orange = class 0.



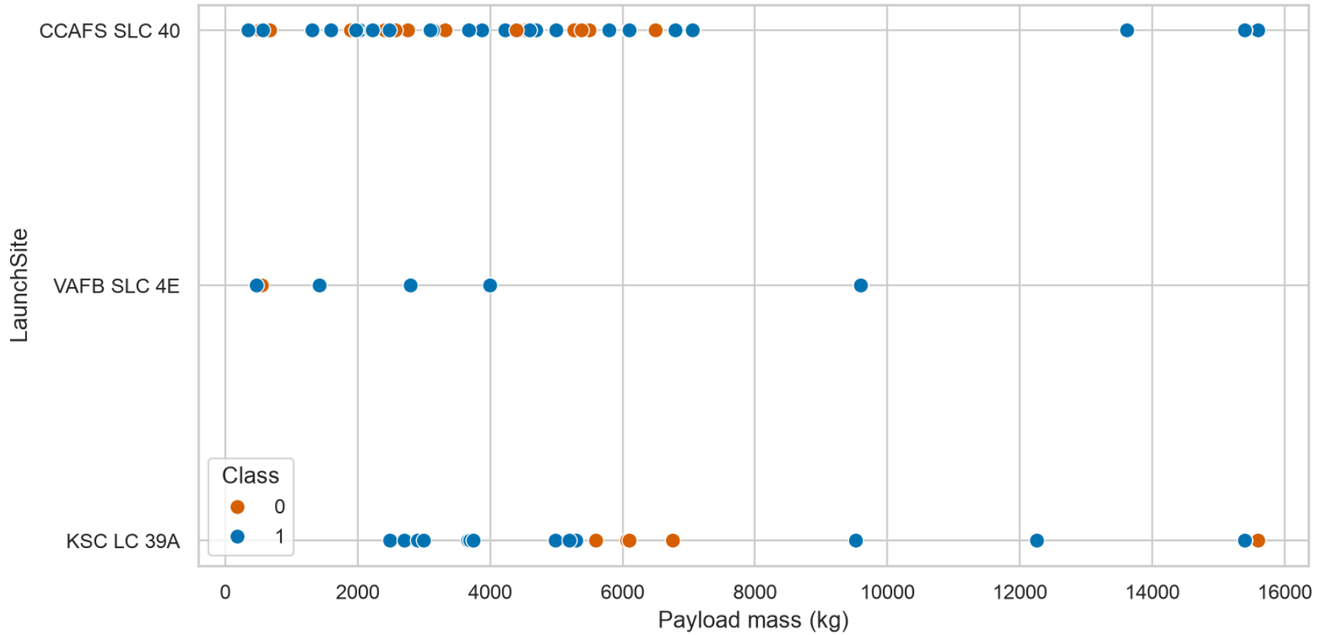
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Payload mass and launch site

Payloads span a wide range at the Florida sites. Successful and unsuccessful outcomes overlap, so payload alone does not separate the classes.

Site codes: 1 = CCAFS SLC 40, 2 = KSC LC 39A, 3 = VAFB SLC 4E.

Blue = class 1. Orange = class 0.



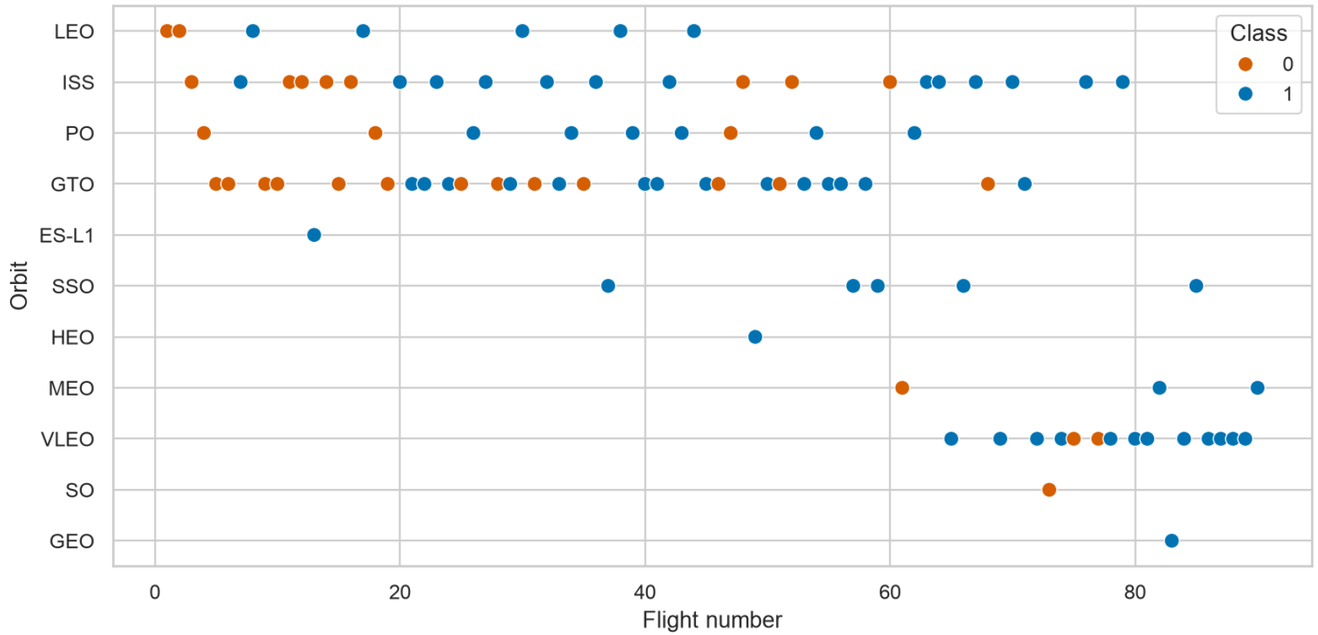
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Flight sequence and orbit

Later VLEO flights mostly show successful landing outcomes. Orbit and launch time are associated in this sample.

Orbit codes 1-11: LEO, ISS, PO, GTO, ES-L1, SSO, HEO, MEO, VLEO, SO, GEO.

Blue = class 1. Orange = class 0.



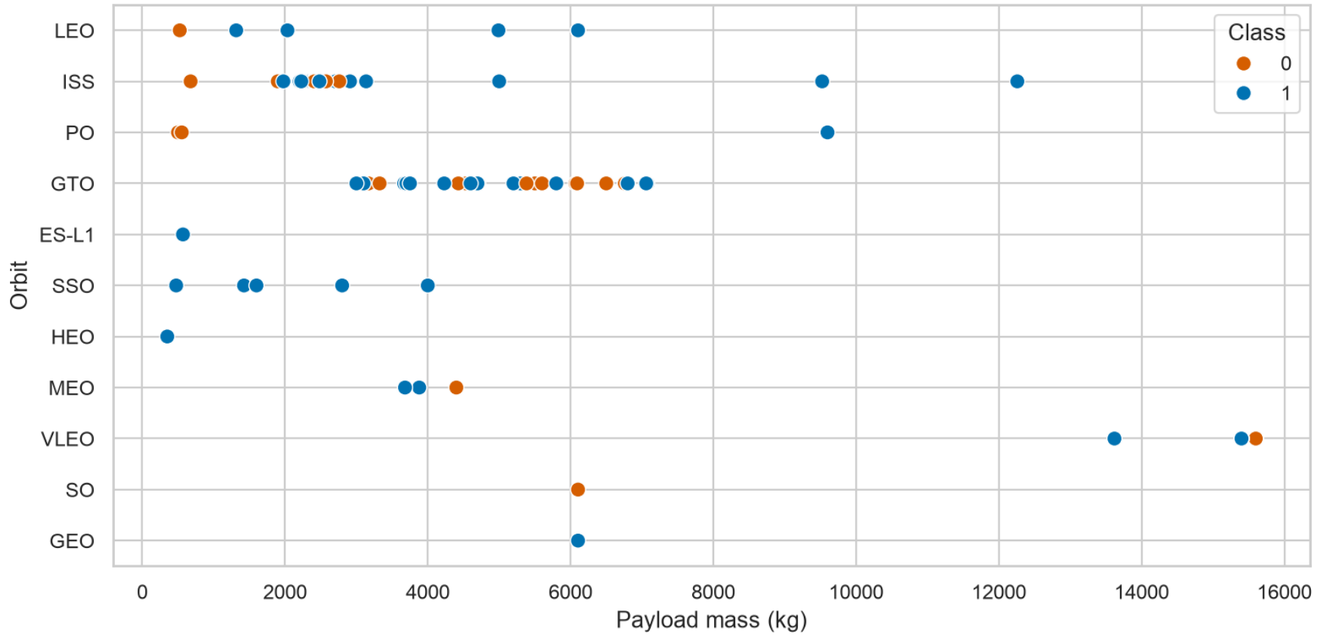
https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

Payload mass and orbit

Payload distributions differ across orbit groups. GTO includes both classes over a broad mass range.

Orbit codes follow first appearance in the dataset: LEO, ISS, PO, GTO, ES-L1, SSO, HEO, MEO, VLEO, SO, GEO.

Blue = class 1. Orange = class 0.

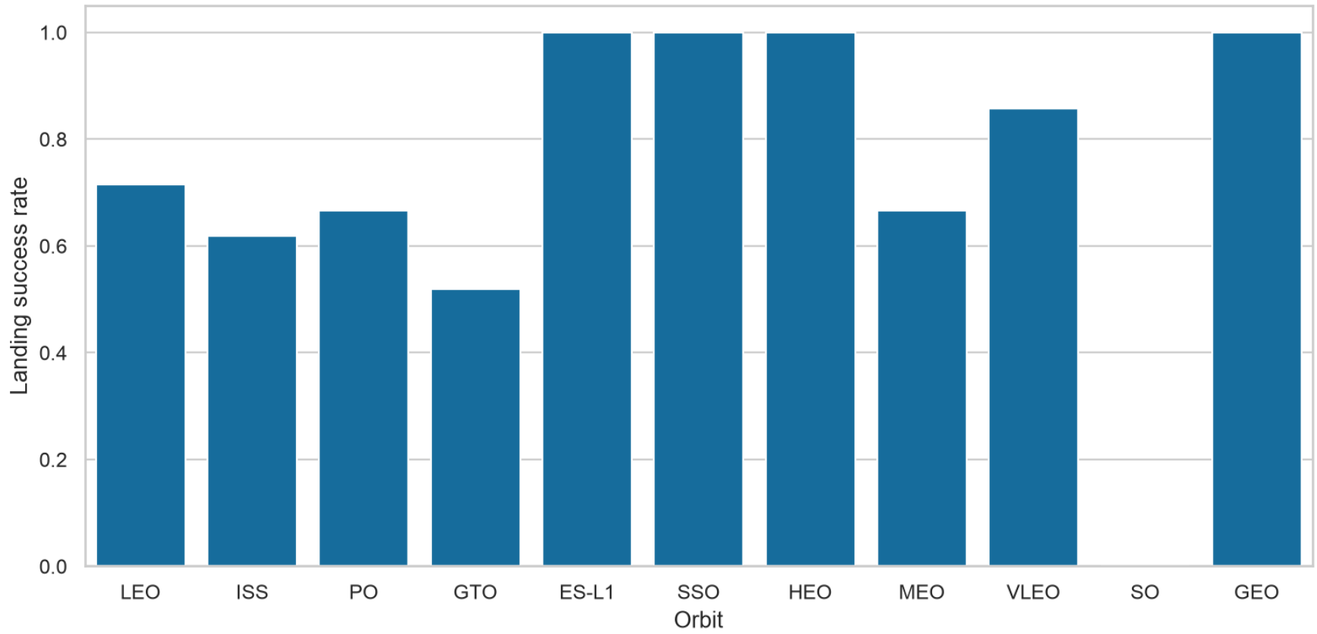


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Landing success by orbit

VLEO: 85.7% across 14 launches.
ISS: 61.9% across 21 launches.
GTO: 51.9% across 27 launches.

Several groups have 100% success but only one observation. SSO has five. These groups do not establish a reliable ranking.



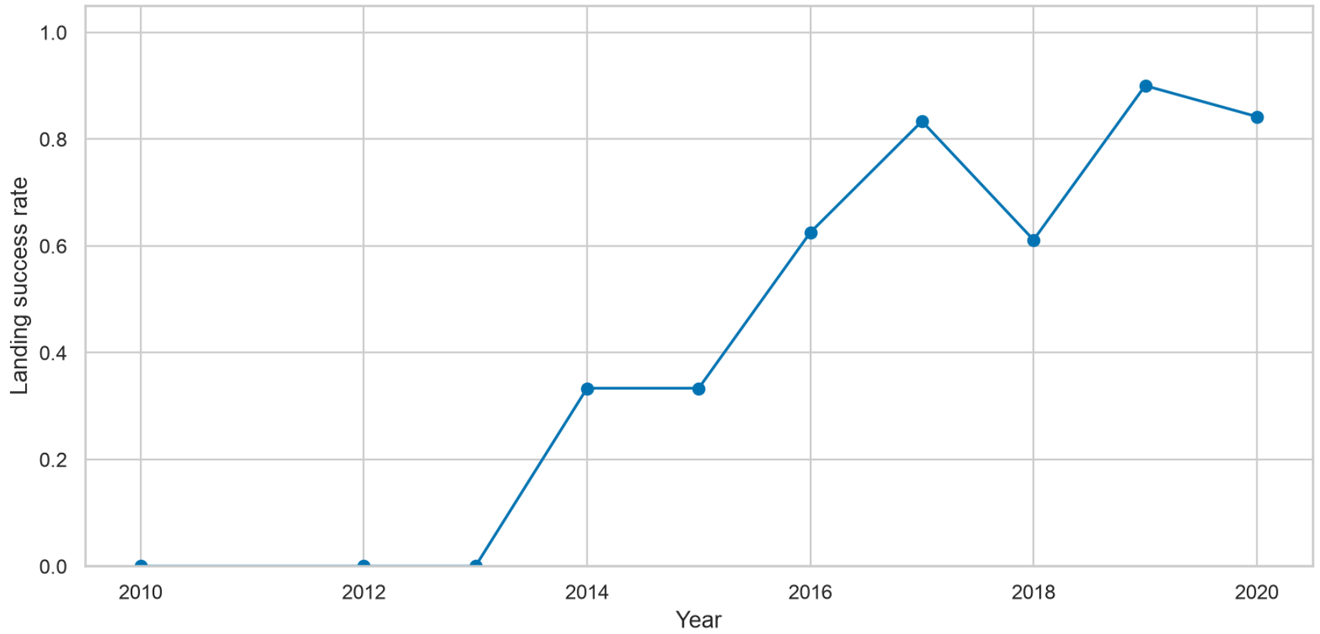
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Yearly landing trend

The observed rate rose from zero in early years to 90.0% in 2019 and 84.2% in 2020.

The change is uneven: 2018 had 11 successes in 18 launches.

Yearly sample sizes vary. The 2020 sample stops on 13 Nov. This pattern supports further study of time and hardware effects.



https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

EDA with SQL methodology

Load the official Spacex.csv into SQLite and trim string whitespace.

Use DISTINCT and LIKE for site queries, SUM and AVG for payloads, MIN for the first ground landing, and a subquery for maximum payload.

GROUP BY summarizes mission outcomes. Date filters and descending counts support temporal analysis.

All 11 executed queries and their result tables appear in the notebook.

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

SQL results: launch sites

The SQL snapshot retains four historical site labels.

The LIKE 'CCA%' query returns the first five Cape Canaveral records. Its earliest listed date is 4 Jun 2010.

Historical LC-40 and SLC-40 labels remain separate here, as in the source.

Launch site	Records
CCAFS LC-40	26
CCAFS SLC-40	34
KSC LC-39A	25
VAFB SLC-4E	16

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SQL results: payload and first landing

NASA (CRS) payload total: 45,596 kg.

Average payload for F9 v1.1 boosters: 2,534.67 kg.

First successful ground-pad landing in the supplied SQL data: 22 Dec 2015.

These values come from SUM, AVG and MIN queries, with customer, booster and landing-outcome filters.

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

SQL results: drone-ship payload filter

Four booster records have successful drone-ship landings and payload mass strictly greater than 4,000 kg and strictly less than 6,000 kg.

The filter uses the explicit historical label Success (drone ship). Later generic Success labels remain outside this query.

Booster	Payload (kg)
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

SQL results: mission outcomes

The 101-row SQL dataset contains 99 ordinary successes, one success with unclear payload status and one in-flight failure.

These are mission outcomes. They cannot substitute for the first-stage landing class used by the predictive model.

Mission outcome	Records
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

SQL results: maximum payload boosters

The maximum payload mass is 15,600 kg. Twelve records share that maximum.

The subquery compares each record with SELECT MAX(PAYLOAD_MASS__KG_). All matching boosters appear at right.

Booster	Payload (kg)
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

SQL results: failures and timing

The explicit drone-ship failures in 2015 occur in January and April, both at CCAFS LC-40.

January: F9 v1.1 B1012.
April: F9 v1.1 B1015.

The table ranks landing outcomes between 4 Jun 2010 and 20 Mar 2017, inclusive. No attempt is the largest group.

Landing outcome	Records
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

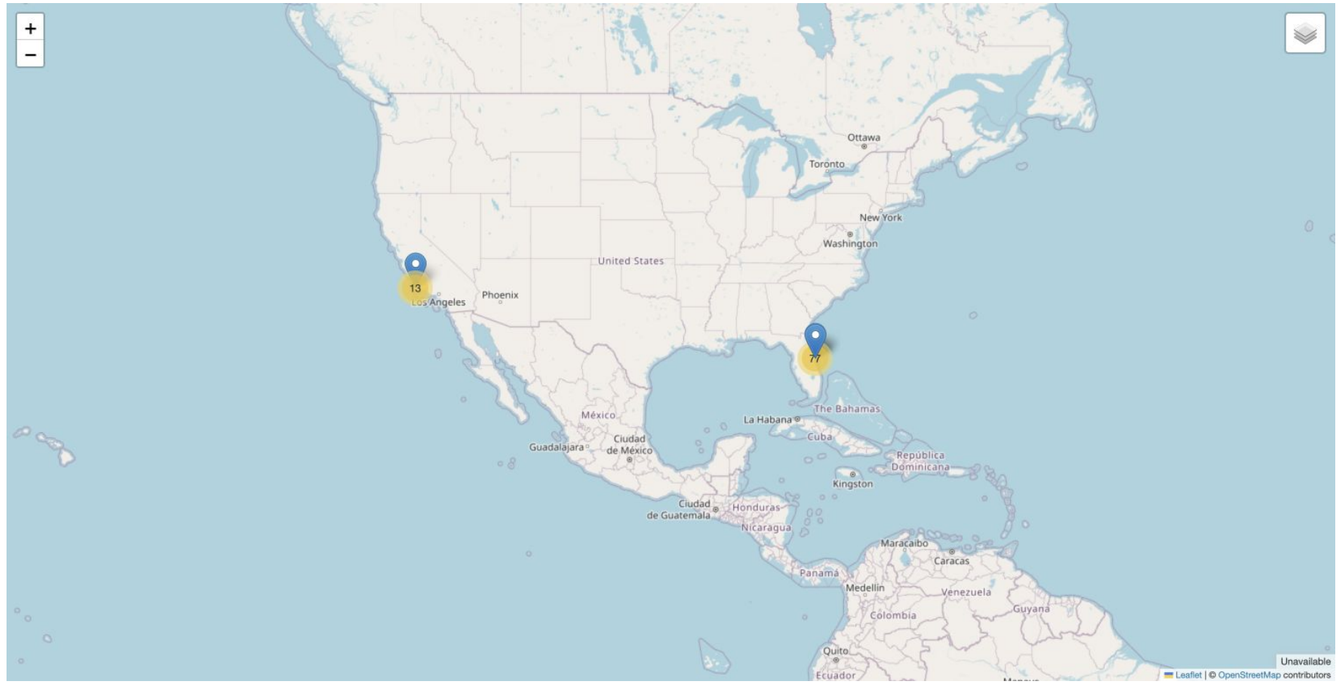
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Interactive visual analytics with Folium

Site markers locate the three API launch sites on the US coasts. Clusters aggregate 90 launch records at shared coordinates.

Hover and popup text identify sites. Cluster markers prevent repeated coordinates from hiding all records.

The map uses OpenStreetMap tiles and the coordinates in the official dataset.



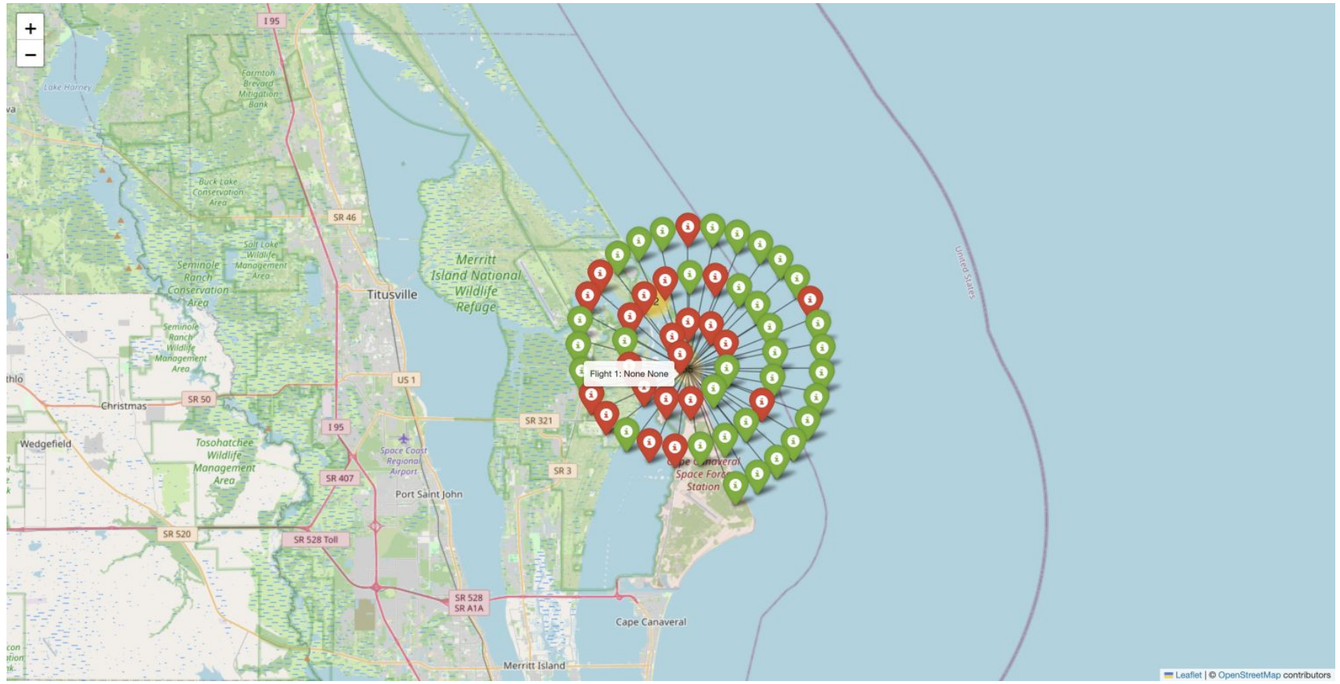
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Folium launch records

Clicking the CCAFS cluster expands its 55 launch records. Green icons mark successful first-stage outcomes and red icons mark the other class.

The expanded icons spread around their shared launch location for readability. Their displayed offsets are artificial.

A hover tooltip identifies each flight and outcome.



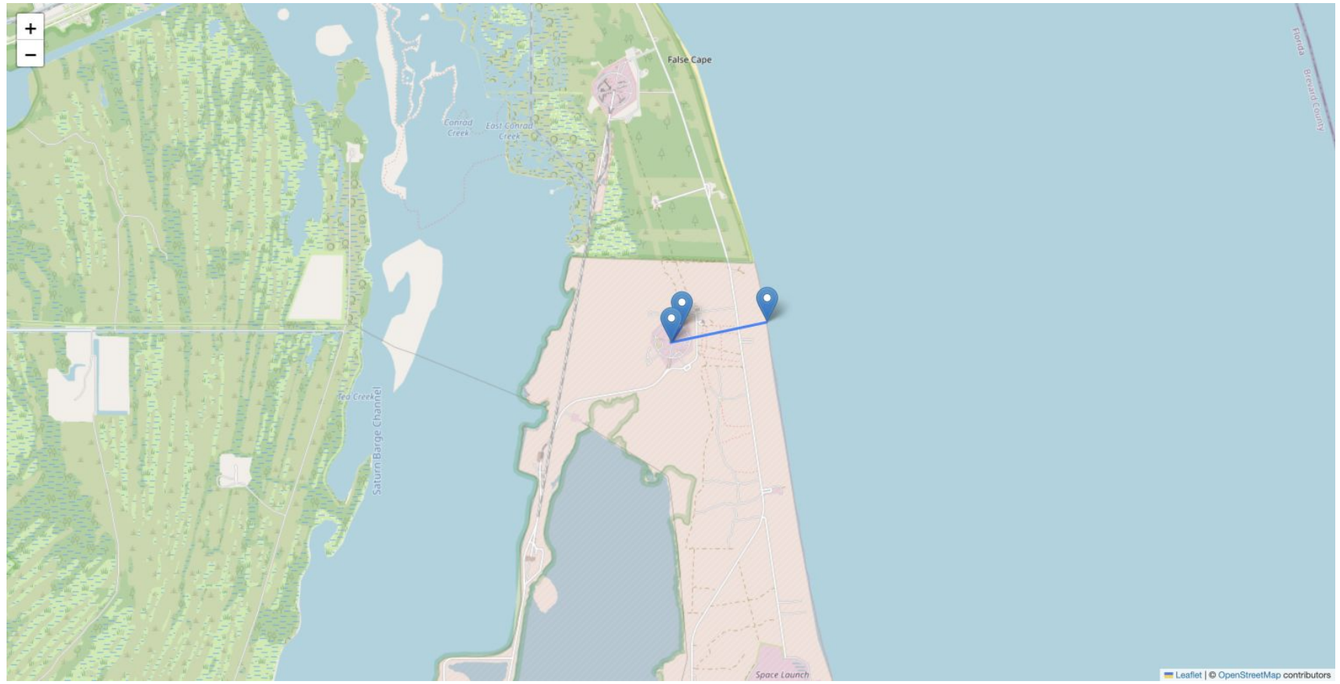
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Folium proximity analysis

The CCAFS SLC 40 inset connects the launchpad to two approximate points selected from the basemap.

Coastline point: about 0.937 km.
Nearby road point: about 0.182 km.

Geodesic distances illustrate proximity to coast and access infrastructure. They are not surveyed distances or safety-clearance estimates.



https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

Interactive visual analytics with Dash

The 56-row dashboard snapshot supplies four historical site labels.

A searchable dropdown selects all sites or one site. The pie chart switches between successful-launch counts by site and outcome counts within one site.

A payload range slider filters the scatter plot. Point colors identify booster versions. Two callback inputs update the figure.

Both controls were exercised in the running local application.

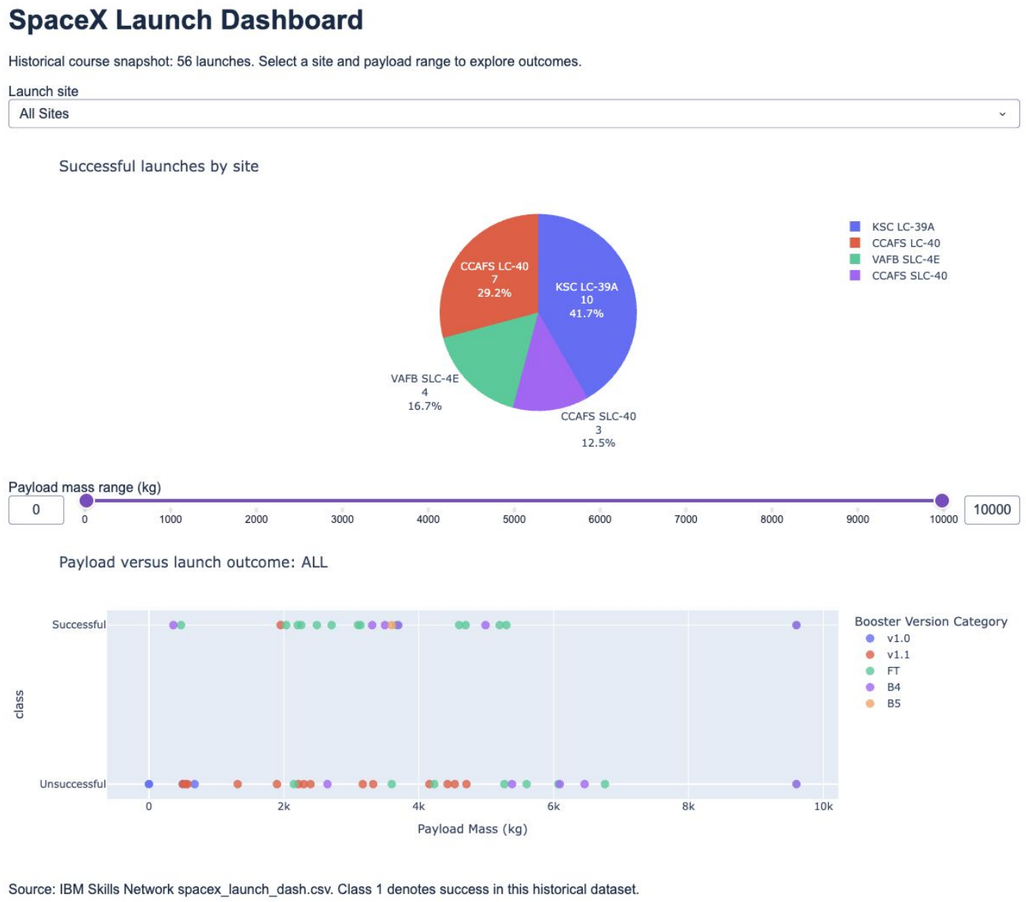
https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/spacex_dash_app.py

Dash results: all launch sites

KSC LC-39A contributes the largest count of successful launches in the dashboard snapshot: 10 of 24 successes (41.7%).

CCAFS LC-40 contributes 7, VAFB SLC-4E contributes 4 and CCAFS SLC-40 contributes 3.

The full dashboard combines the pie chart, dropdown and payload scatter.



Source: IBM Skills Network spacex_launch_dash.csv. Class 1 denotes success in this historical dataset.

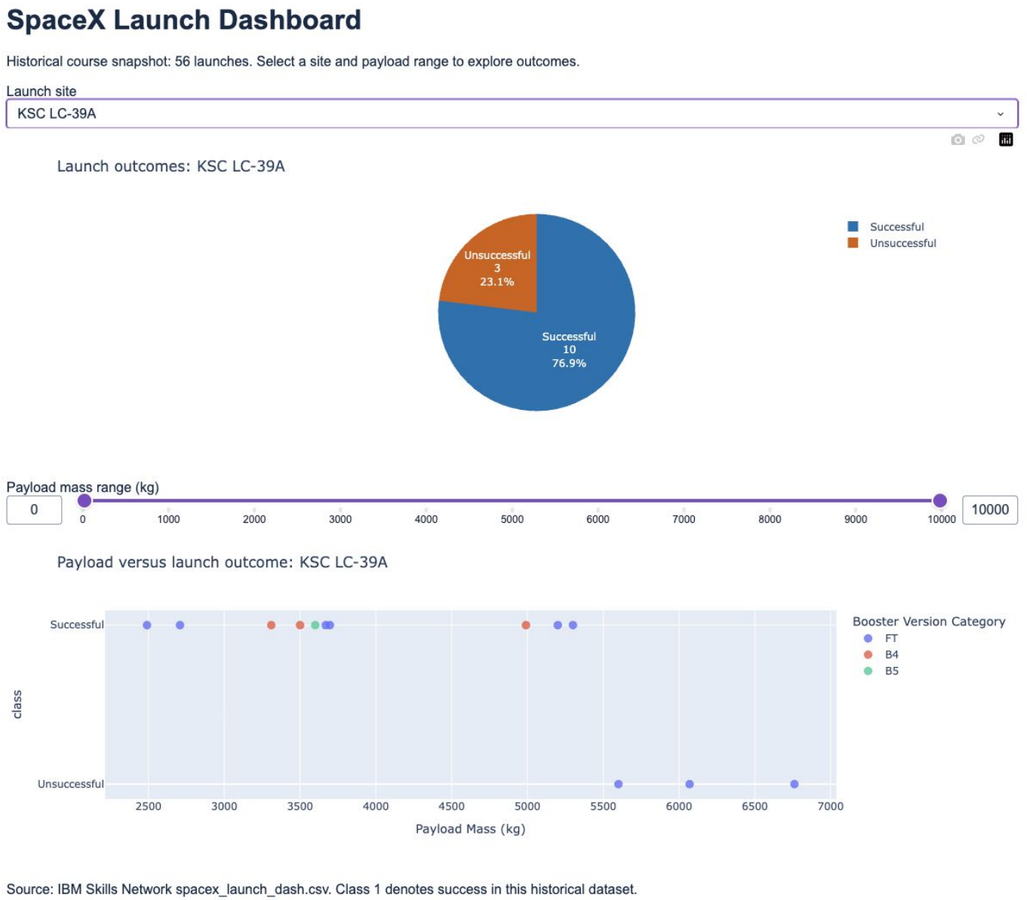
https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/spacex_dash_app.py

Dash results: KSC LC-39A

Selecting KSC LC-39A changes the pie to 10 successes and 3 unsuccessful launches, for a 76.9% observed success rate.

This is the highest site rate in the 56-row dashboard snapshot. It is separate from the 90-row API analysis.

The scatter plot now contains only the selected site.



Source: IBM Skills Network spacex_launch_dash.csv. Class 1 denotes success in this historical dataset.

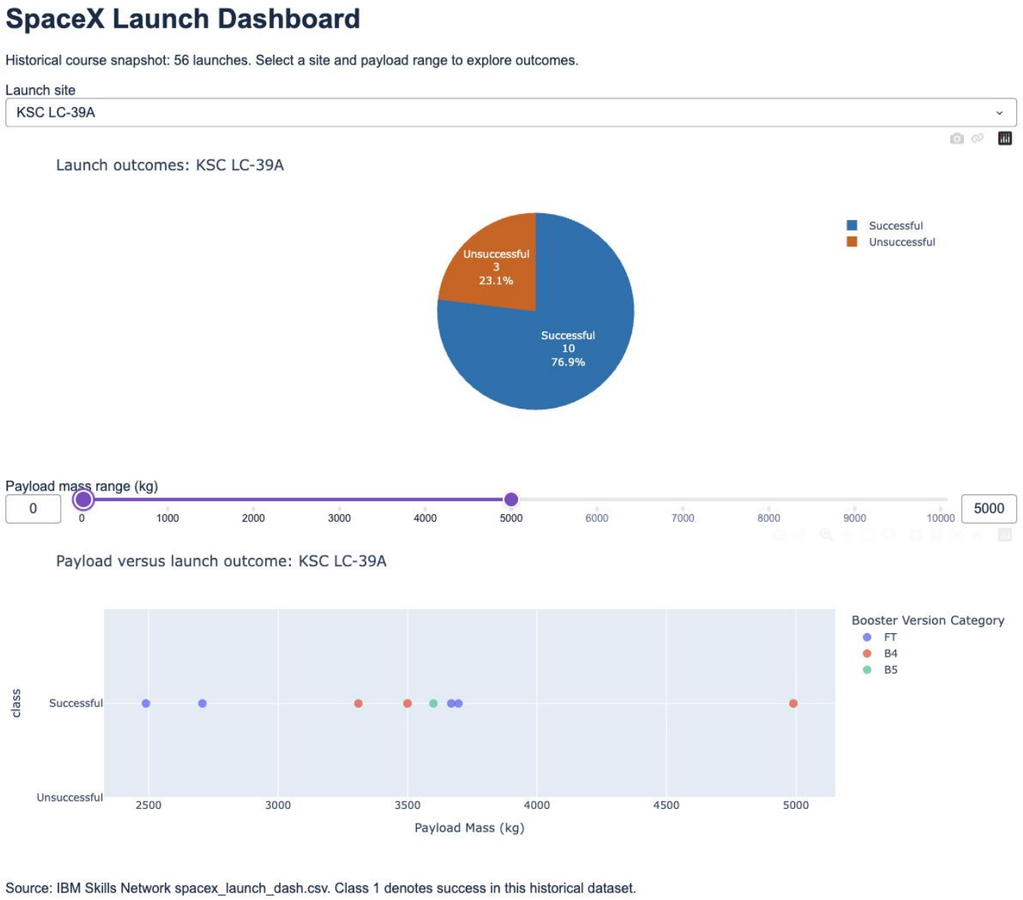
https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/spacex_dash_app.py

Dash results: payload and booster

A 0-5,000 kg range further filters KSC LC-39A points.

Across all dashboard sites, 2,000-4,000 kg has the highest observed success rate among the specified 2,000 kg bins: 13/21 (61.9%). The 6,000-8,000 kg bin has 0/4.

B5 shows 1/1 success, while FT shows 16/24. The single B5 observation cannot establish superiority.



Source: IBM Skills Network spacex_launch_dash.csv. Class 1 denotes success in this historical dataset.

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/spacex_dash_app.py

Predictive analysis methodology

Train/test split: 72/18 rows, stratified, random_state=2.

Five-fold stratified cross-validation tunes logistic regression, SVM, a decision tree and KNN on training data.

Each pipeline imputes missing values, encodes categories and scales numerical features inside each fold.

Select the best mean CV accuracy, then report held-out performance. Upstream payload imputation limits evaluation. ReusedCount may reflect later reuse. Reconstruct features available at launch time before prospective prediction.

https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

Predictive analysis results

The decision tree has the highest mean CV accuracy: 90.4%, with a fold standard deviation of 5.2 percentage points.

Its selected depth is 3 and minimum leaf size is 4. Test accuracy is 83.3%, with balanced accuracy of 75.0%.

The always-successful baseline is 66.7% on this test split.

Model	CV accuracy	Test accuracy
Decision tree	90.4%	83.3%
KNN	89.1%	66.7%
SVM	86.3%	77.8%
Logistic regression	83.6%	83.3%

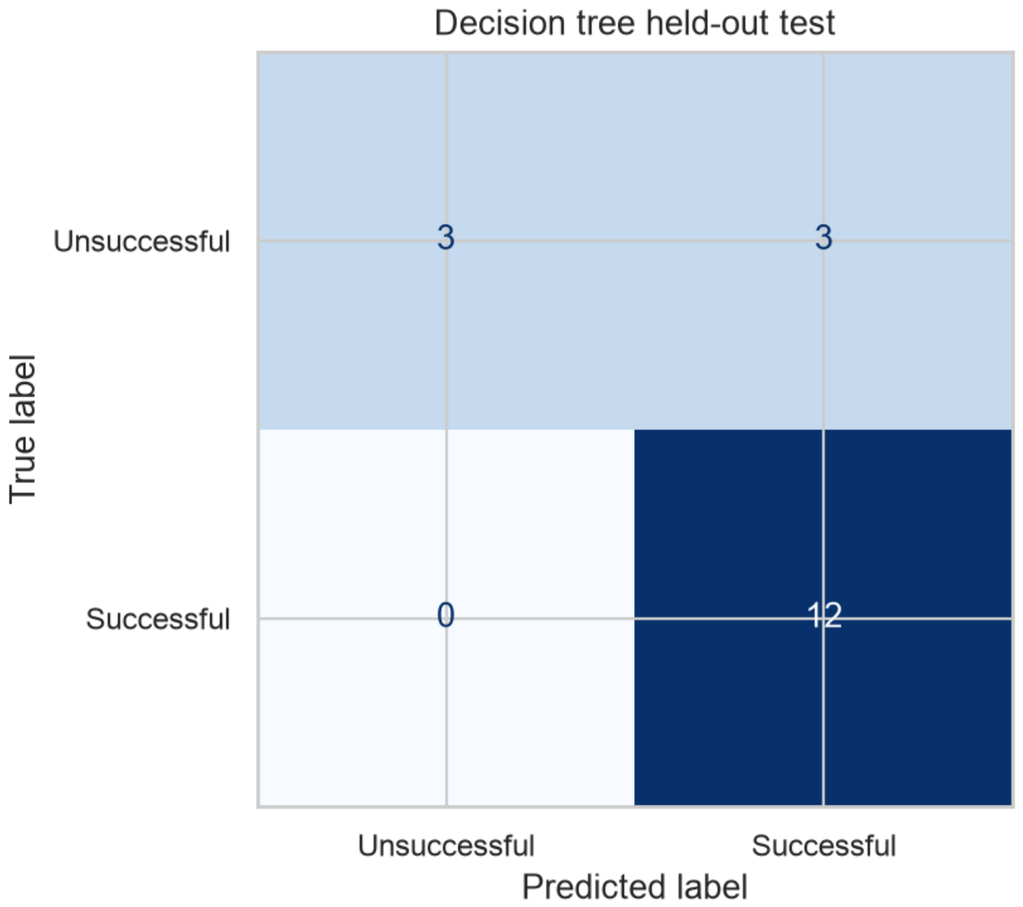
https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

Confusion matrix and uncertainty

On 18 held-out rows, the tree correctly identifies 3 of 6 unsuccessful outcomes and all 12 successful outcomes.

Three unsuccessful outcomes receive successful predictions. This error matters for reuse planning.

One test error changes accuracy by 5.6 percentage points. A larger chronological holdout and calibration are needed.



https://github.com/Zeppelin567/ibm-spacex-capstone/blob/main/SpaceX_Capstone_Analysis.ipynb

Conclusion and next steps

Observed landing success improved over time. Payload, orbit, site and hardware are associated with outcomes, but the samples do not identify causal effects.

The decision tree provides a transparent starting point. A chronological evaluation should test whether its advantage survives changes in launch operations.

For economic decisions, distinguish ocean landing success from recovery suitable for reuse. Add weather, mission constraints and refurbishment data.

Publish the reproducible notebook, data provenance, dashboard and maps together.

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Project files and sources

https://github.com/Zeppelin567/ibm-spacex-capstone

SpaceX_Capstone_Analysis.ipynb contains executed collection, wrangling, EDA, SQL, mapping and modeling.
spacex_dash_app.py runs the interactive dashboard.
CSV, JSON and HTML snapshots preserve the inputs.

AI assistance supported preparation of the code and presentation. All reported calculations come from executed analysis.

Official IBM datasets and lab instructions. Wikipedia revision oldid=1027686922. Folium basemaps © OpenStreetMap contributors.